



Lect 02

Introduce active high pass filters

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Analog Electronics Lab



Assignments

Objective: Design a three-pole high-pass Butterworth active filter with a cutoff frequency at $f_{3dB} = 30$ kHz and a unity gain magnitude at high frequency.

Specifications: The circuit with the configuration shown in Figure below is to be designed such that the bandwidth is 30 kHz.

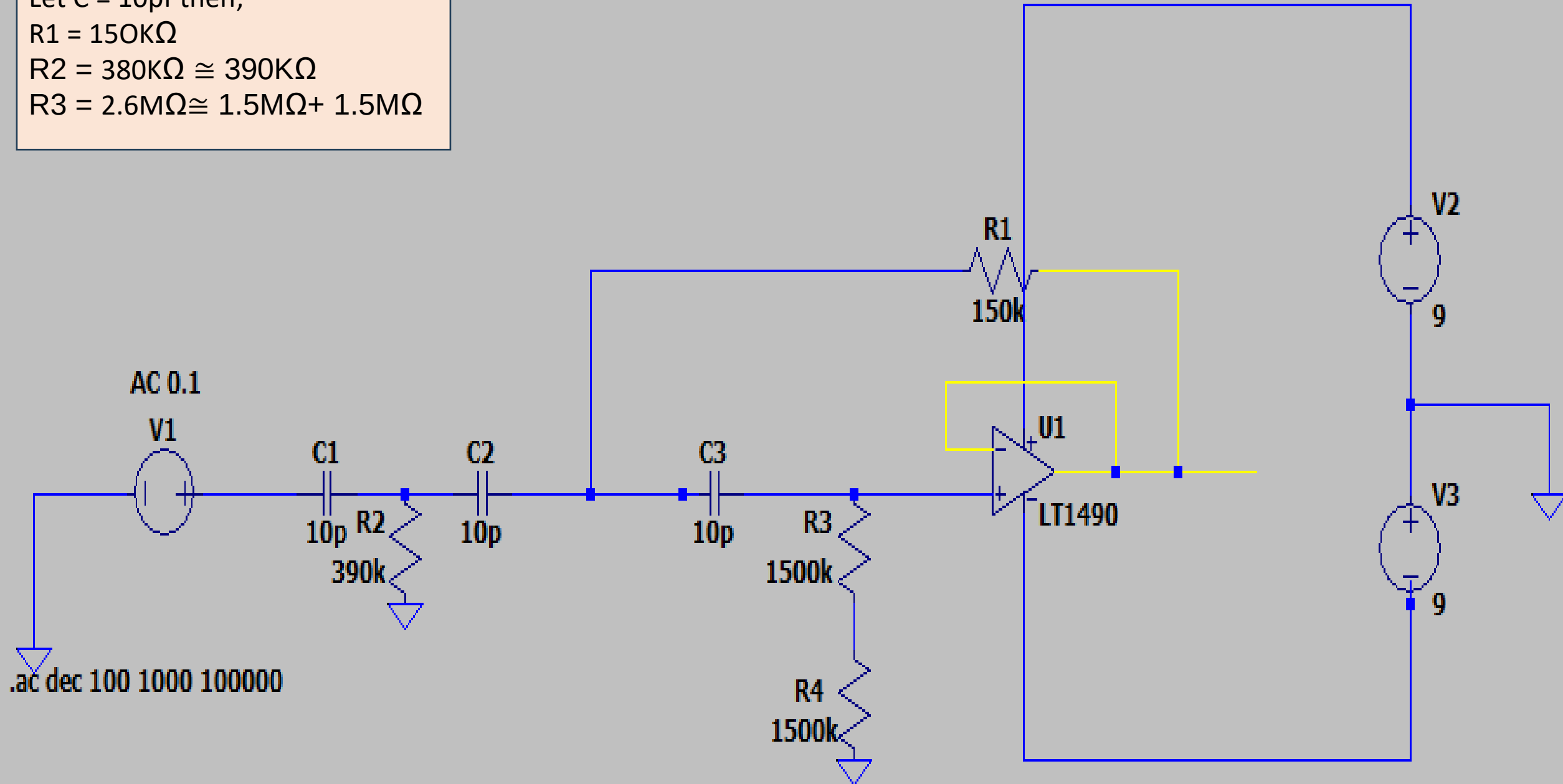
Choices: An ideal op-amp is available and standard-valued resistors and capacitors must be used.

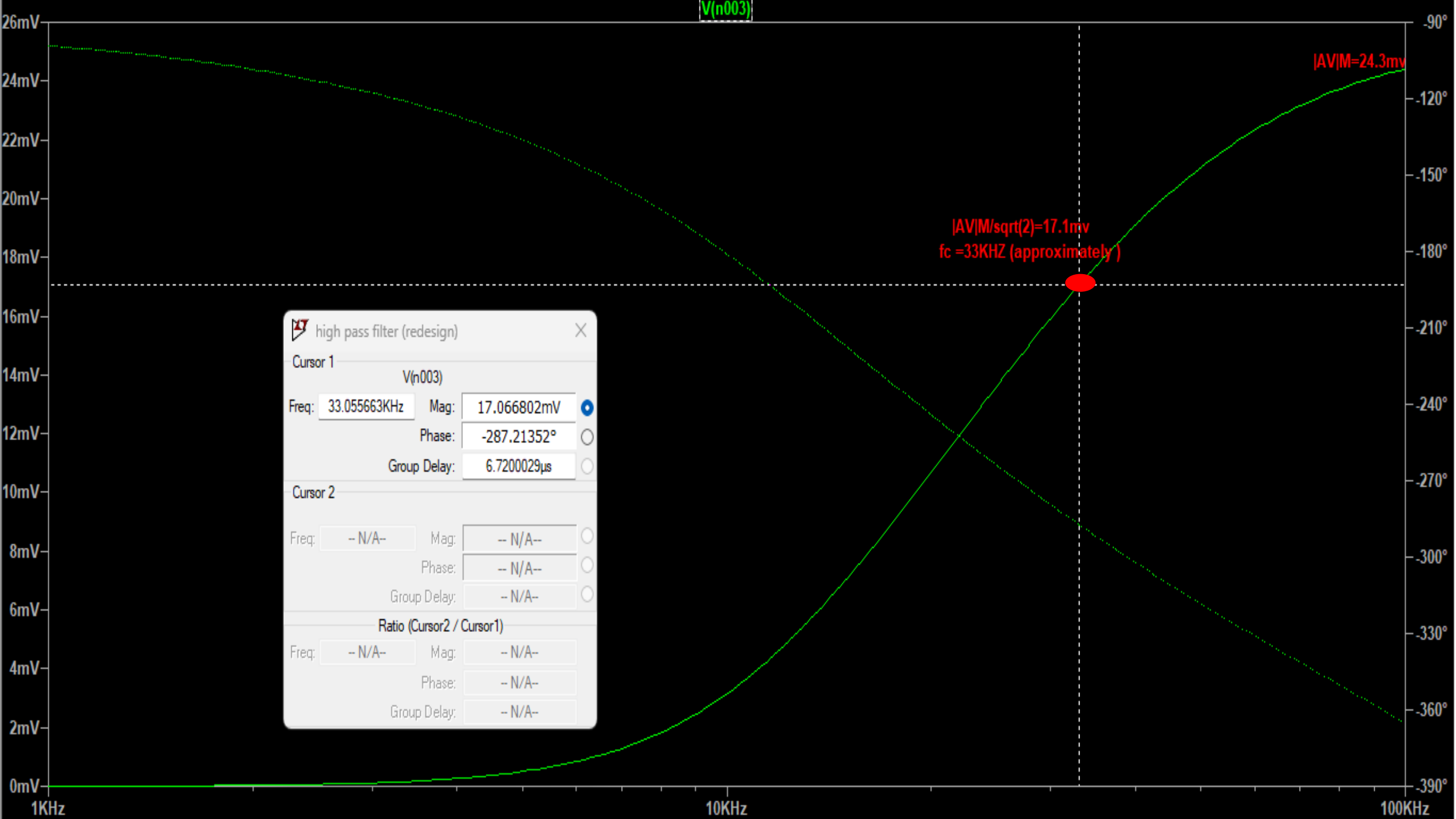
Bode plot : Determine the magnitude of the gain at (i) $f = 25$ kHz, (ii) $f = 30$ kHz, and (iii) $f = 40$ kHz.

computer simulation LTSPICE : Using a computer simulation, plot the magnitude of the voltage transfer function versus frequency and compare these results with your plot

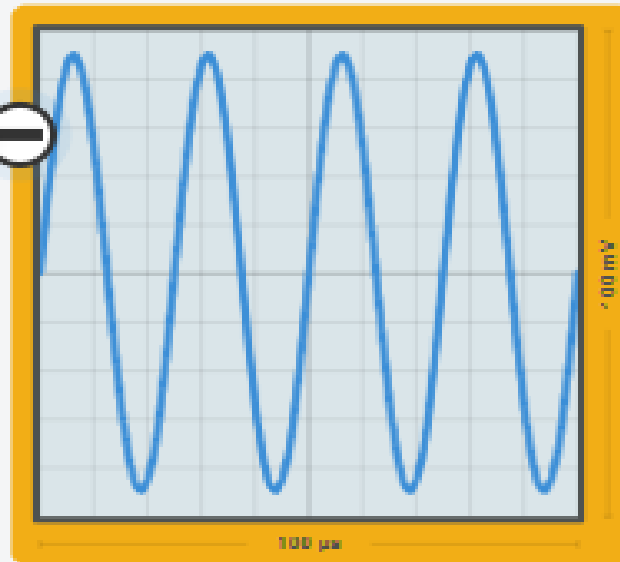


Let $C = 10\text{pf}$ then;
 $R1 = 150\text{K}\Omega$
 $R2 = 380\text{K}\Omega \cong 390\text{K}\Omega$
 $R3 = 2.6\text{M}\Omega \cong 1.5\text{M}\Omega + 1.5\text{M}\Omega$

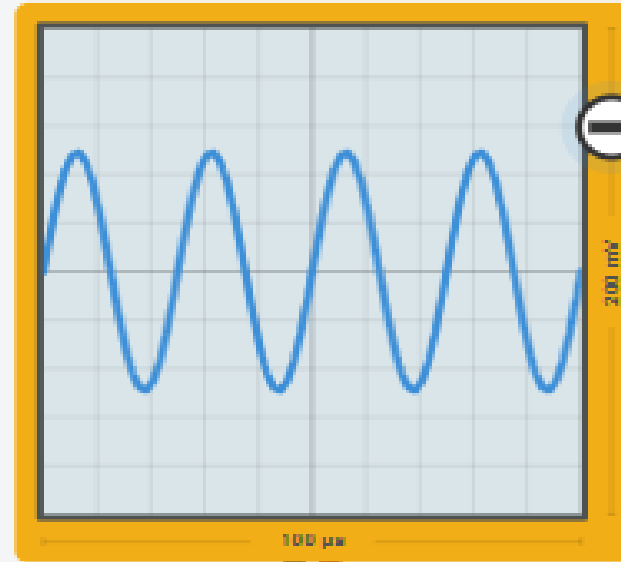




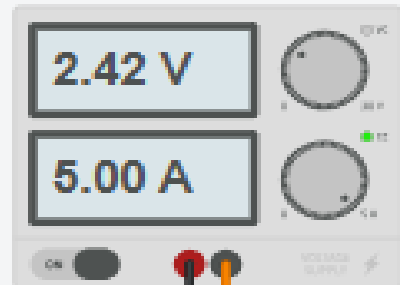
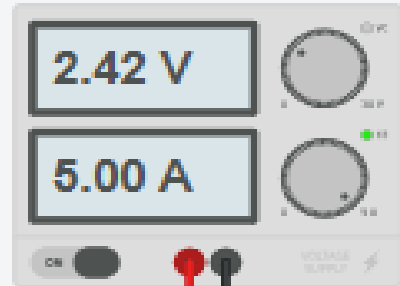
Output Signal



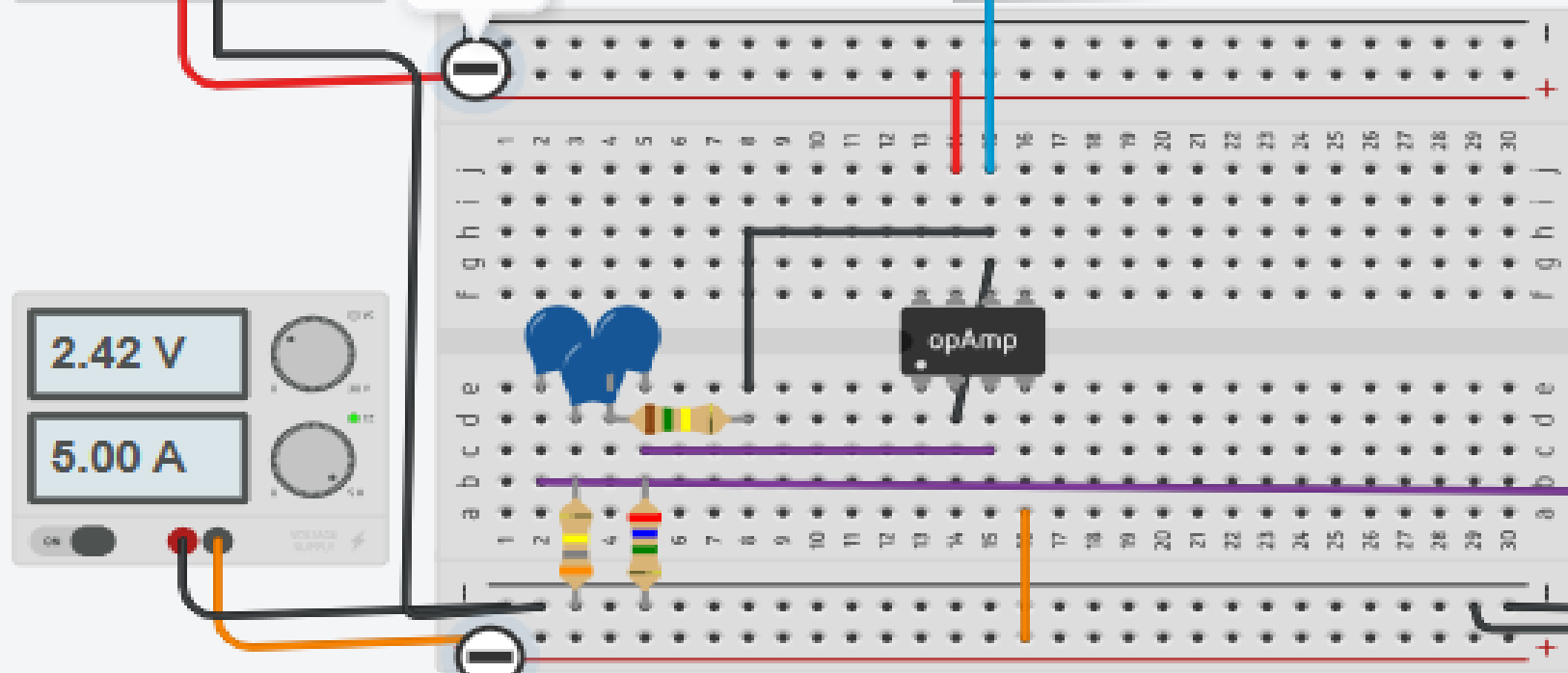
Input Signal



+9v



-9v



opAmp

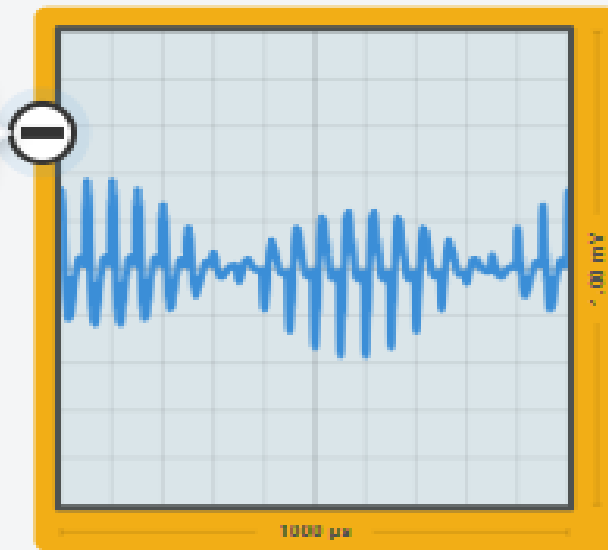
40.0 kHz

100 mV

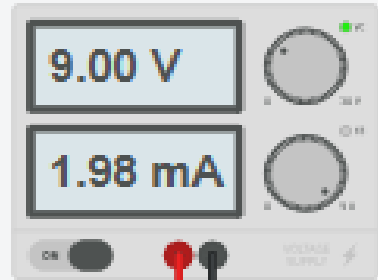
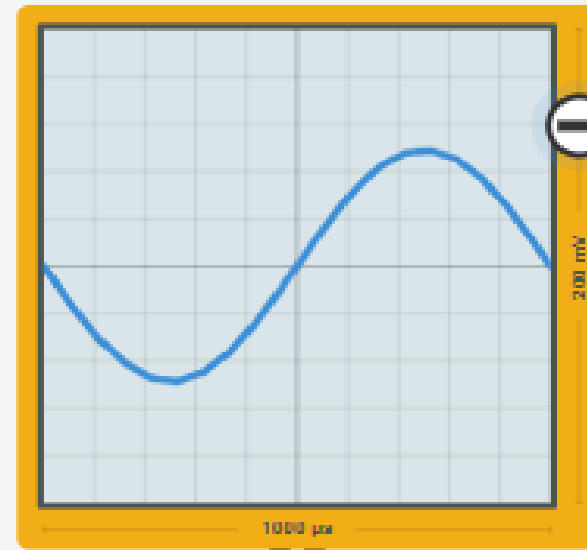
0.00 V



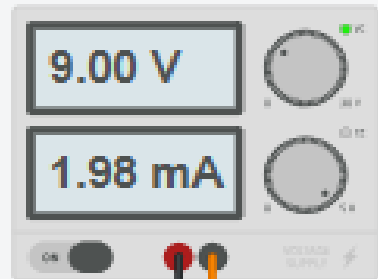
Output Signal



Input Signal



+9v



-9v

